

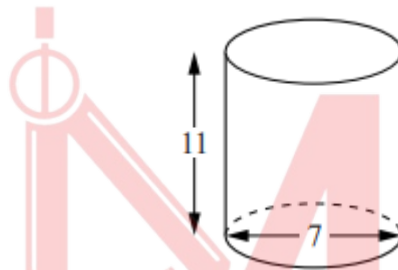
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Worksheet 2d

Cylinder

1. W12/qp21/q7(b)

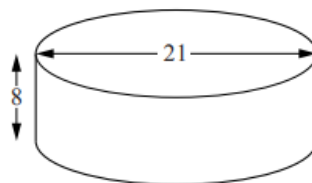
A closed cylindrical tin is 11 cm high and the base has a diameter of 7 cm.



- i. Calculate the volume of this tin. [2]
- ii. Calculate the total external surface area of this tin. [3]
- iii. In addition to the surface area, a closed tin requires an extra 30 cm^2 of metal to allow the top, bottom and side to be joined together. Calculate the area of metal required for 30 000 closed tins. Give your answer in square metres. [2]

2. W12/qp22/q7(a)

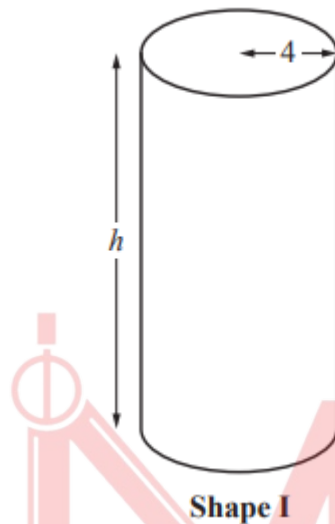
A cylindrical, open container has a diameter of 21 cm and height of 8 cm.



- i. Calculate the total external surface area of this container. [3]
- ii. A manufacturer receives an order for 30 000 containers. He needs an extra 150 cm^2 of material for each container to cover wastage. Calculate the area of material needed to make these containers. Give your answer in square metres. [2]

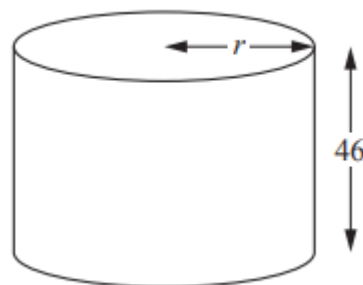
3. S13/qp21/q9(a)

Shape I is a cylinder with radius 4cm and height h cm. The volume of Shape I is 1500cm^3 .



- i. Find h . [2]
- ii. Shape I is made by pouring liquid into a mould at a rate of 0.9 litres per minute. Find the number of seconds it takes to pour this liquid into the mould. [1]

4. S13/qp22/q12(a)

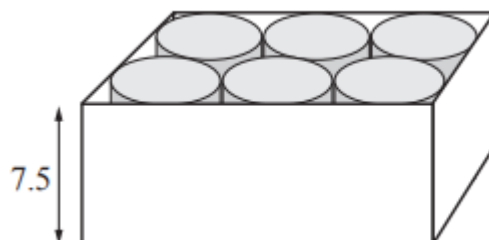


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A cylindrical tank of height 46cm and radius r cm has a capacity of 70 litres. Find the radius correct to the nearest centimetre. [3]

5. S14/qp21/q6(a)

- a. A candle is in the shape of a cylinder of radius 1.6cm and height 7.5 cm.
- b.
- Calculate the volume of the candle. [2]
 - Six of these candles are packed into a box of height 7.5cm as shown.



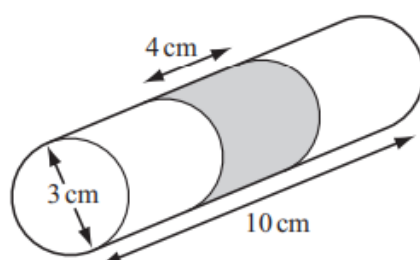
- Find the length and width of the box. [1]
- Calculate the volume of empty space in the box. [2]

6. S14/qp33/q5(b) 0580

The volume, V , of a cylinder with radius r and height h is $V = \pi r^2 h$.

- Calculate the volume of a cylinder with radius 3cm and height 12cm. [2]
- Ravi puts 150 identical marbles in the cylinder. He fills the cylinder to the top with 160cm^3 of water. Find the volume of one marble. Give your answer correct to 2 significant figures. [4]
- Make r the subject of the formula $V = \pi r^2 h$. [2]

7. W14/qp33/q8(b) 0580



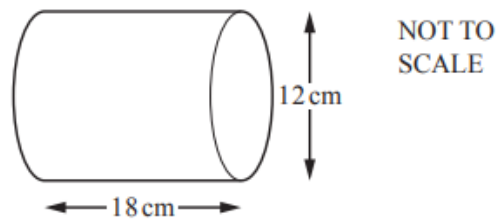
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Another box of sweets is in the shape of a cylinder. The cylinder has diameter 3cm and length 10cm.

- Calculate the volume of the cylinder. [3]
- A label of width 4cm fits around the cylinder with no overlap. Calculate the area of the label. [3]

8. S15/qp13/q20(a) 0580

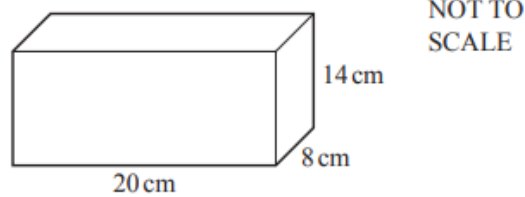
- a. A cylinder has diameter 12 cm and length 18 cm.



Calculate the volume of the cylinder.

[2]

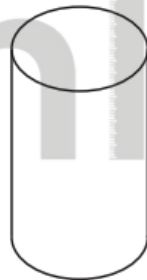
- b.



- i. Calculate the surface area of this cuboid. [3]
- ii. Write your answer to part (b)(i) in square millimetres. [1]

9. W15/qp21/q11(b)

A cylinder has a height of 70cm and a volume of $0.1m^3$. Calculate the radius of the cylinder, giving your answer in centimetres. [4]



10. W15/qp22/q8(c)

A dairy produces cylindrical cheeses, each with a volume of $800 cm^3$. The height h cm and the radius r cm can vary.

- i. Express h in terms of r . [1]
- ii. What happens to the height if the radius is doubled? [1]

Answer Key

1. W12/qp21/q7(b)

(b) (i) 423 to 424 (ii) 319 (iii) 1050	2 M1 for $\pi \times r^2 \times 11$. 3 M2 for $2\pi r^2 + 2\pi r \cdot 11$ or M1 for either of these 2ft M1 for Figs (their 319 + 30) $\times$ 3 B1 for $\div 10^4$
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2. W12/qp22/q7(a)

7 (a) (i) 874 (ii) 3070	3 M2 for (2) $\pi r^2 + 2\pi r \times 8$ or M1 for either (2) πr^2 or $2\pi rh$ 2ft M1 for Figs [(their 874 + 150) $\times$ 3] or B1 for $\div 10^4$
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3. S13/qp21/q9(a)

9 (a) (i) $h = 29.8$ to 29.85 (ii) 100	2 M1 for $\pi \times 4^2 \times h (=1500)$ 1
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4. S13/qp22/q12(a)

12 (a) $r = 22$ cao	3 B1 for 70000 soi M1 for $\pi \times r^2 \times \text{figs}46$ (only term)
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5. S14/qp21/q6(a)

6 (a) (i) 60.28 to 60.35 (ii) (a) length 9.6, width 6.4 (b) 98.7 to 99.2	2 M1 for $\pi \times 1.6^2 \times 7.5$ 1 Condone reversed 2 M1 for 'their $9.6 \times 6.4 \times 7.5 - 6 \times$ 'their 60.3' Or B1 for 460.8, or 361.68 to 362.1
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6. S14/qp33/q5(b) 0580

(b) (i)	339	2	M1 $\pi \times 3^2 \times 12$
(ii)	1.2 cao	4	M2 FT for $\frac{\text{their } 339 - 160}{150}$ soi or M1 FT for $\text{their } 339 - 160$ soi A1 for 1.19... If A0 scored then B1 for correct rounding of their 3 sig fig or more answer.
(iii)	$r = \sqrt{\frac{v}{\pi h}}$	2	M1 for $r^2 = \frac{v}{\pi h}$

7. W14/qp33/q8(b) 0580

(b) (i)	339	2	M1 $\pi \times 3^2 \times 12$
(ii)	1.2 cao	4	M2 FT for $\frac{\text{their } 339 - 160}{150}$ soi or M1 FT for $\text{their } 339 - 160$ soi A1 for 1.19... If A0 scored then B1 for correct rounding of their 3 sig fig or more answer.
(iii)	$r = \sqrt{\frac{v}{\pi h}}$	2	M1 for $r^2 = \frac{v}{\pi h}$

8. S15/qp13/q20(a) 0580

20 (a)	2040 or 2036 or 2035.7 to 2036.1	2	M1 for $\pi \times 6^2 \times 18$ If zero, SC1 for answer of 8140 (after use of diameter)
(b)(i)	1104	3	M2 for $2 \times (20 \times 14 + 20 \times 8 + 8 \times 14)$ oe or M1 for $20 \times 14 + 20 \times 8 + 8 \times 14$
(ii)	$\text{their (b)(i)} \times 100$	1FT	

9. W15/qp21/q11(b)

(b)	21.3[2...]	4	B1 for correct change of units soi M1 for use of $\pi \times r^2 \times 0.7 = 0.1$ M1 for $r^2 = \frac{0.1}{0.7 \times \pi}$ soi
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10. W15/qp22/q8(c)

(c) (i)	$(h =) \frac{800}{\pi \cdot^2}$	1	
(ii)	h is divided by 4 oe	1	